

NIZHNEUSINSKOYE, Russian Federation

WMO: 360910

Lat: 52.257N

Lon: 93.013E

Elev: 2172

StdP: 13.58

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.0	(c) -34.3	(d) -44.1	(e) 0.5	(f) -38.6	(g) -39.0	(h) 0.6	(i) -33.9	(j) 11.0	(k) -12.8	(l) 6.8	(m) -13.8	(n) 0.4	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.3	(c) 86.4	(d) 63.8	(e) 82.4	(f) 63.3	(g) 78.7	(h) 61.7	(i) 67.0	(j) 79.6	(k) 65.2	(l) 77.3	(m) 63.6	(n) 75.0	(o) 3.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.1	(b) 93.8	(c) 70.9	(d) 61.1	(e) 87.3	(f) 68.9	(g) 59.3	(h) 81.8	(i) 67.5	(j) 32.8	(k) 79.6	(l) 31.4	(m) 77.6	(n) 30.1	(o) 75.2	(p) 75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	11.9	9.1	-43.3	93.1	3.6	4.0	-45.8	96.0	-47.9	98.3	-50.0	100.5	-52.6	103.4
			-43.5	70.2	3.4	3.1	-46.0	72.4	-48.0	74.2	-49.9	75.9	-52.3	78.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	-14.9	-3.4	20.6	38.9	47.8	59.9	64.2	60.1	47.2	31.5	7.6	-8.5		
	DBStd	28.57	8.81	10.82	11.14	7.69	7.03	7.23	4.81	5.46	7.27	8.58	12.03	11.03		
	HDD50	8691	2013	1495	911	337	125	9	0	2	138	573	1273	1815		
	HDD65	13079	2478	1915	1376	782	532	181	72	168	534	1038	1723	2280		
	CDD50	1178	0	0	0	5	58	307	439	315	54	0	0	0		
	CDD65	90	0	0	0	0	0	29	46	15	0	0	0	0		
	CDH74	2064	0	0	0	4	121	790	778	339	32	0	0	0		
	CDH80	627	0	0	0	0	27	280	248	69	3	0	0	0		
(14) Wind	WSAvg	1.8	0.5	0.9	1.8	3.7	4.0	2.8	2.4	1.7	1.8	1.2	0.7	0.4		
(15) Precipitation	PrecAvg	28.5	1.0	1.0	1.5	1.9	2.4	3.1	4.1	4.0	2.7	2.7	2.4	2.0		
	PrecMax	37.6	2.4	3.2	4.3	4.0	5.6	5.1	9.6	8.0	4.3	12.0	5.8	6.1		
	PrecMin	19.3	0.1	0.0	0.2	0.5	0.0	0.6	0.2	2.2	1.4	0.6	0.7	0.1		
	PrecStd	4.4	0.6	0.7	1.0	1.1	1.4	1.2	1.5	1.3	0.8	2.2	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	28.8	54.2	73.2	83.9	92.8	91.8	86.0	78.1	60.7	36.4	20.3		
		MCWB	8.3	25.3	41.2	51.6	58.7	63.7	67.6	66.6	60.2	48.3	32.7	18.7		
	2%	DB	5.1	22.5	47.8	65.8	77.3	86.9	86.2	81.6	72.3	53.9	32.3	15.7		
		MCWB	4.1	19.6	37.7	48.6	55.8	62.9	64.6	64.9	57.6	44.6	29.8	14.8		
	5%	DB	1.8	17.9	43.1	60.8	71.7	82.7	81.8	77.6	67.4	48.6	28.0	11.4		
		MCWB	1.1	15.6	34.7	46.0	53.1	61.8	64.1	62.8	55.3	41.3	25.5	10.6		
	10%	DB	-1.3	13.3	38.1	56.2	66.0	78.0	77.9	73.7	62.3	43.9	23.9	6.7		
		MCWB	-2.1	11.5	31.5	43.5	50.9	60.5	62.8	61.3	52.7	38.8	22.0	5.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	25.6	41.4	52.7	60.0	67.4	69.8	69.3	61.6	49.8	33.7	18.8		
		MCDB	9.9	28.3	52.7	69.9	81.5	81.6	85.5	80.3	74.7	57.9	35.9	19.9		
	2%	WB	4.3	19.6	38.3	49.6	57.2	65.1	67.4	66.5	58.6	45.1	29.6	14.9		
		MCDB	5.0	22.3	47.2	64.5	74.4	80.7	80.2	78.2	70.1	51.8	31.8	15.8		
	5%	WB	1.2	15.6	35.0	46.3	54.5	63.3	65.7	64.2	56.1	42.1	26.1	10.6		
		MCDB	1.8	17.7	42.7	58.8	69.3	78.7	77.2	75.5	65.7	48.0	28.1	11.4		
	10%	WB	-2.0	11.5	31.8	44.1	51.7	61.7	64.3	62.1	53.3	39.0	22.2	6.0		
		MCDB	-1.4	13.2	37.7	55.7	64.5	75.6	74.5	71.4	61.2	43.4	24.1	6.7		
(35) Mean Daily Temperature Range	MDBR		19.9	26.4	29.9	27.4	29.0	29.5	25.3	25.0	27.8	20.5	17.5	16.0		
	5% DB	MCDBR	18.0	26.0	30.4	36.8	42.1	39.7	35.2	34.0	36.8	29.9	17.1	14.3		
		MCWBR	17.0	22.6	20.2	20.8	22.2	17.3	16.2	18.1	21.7	19.4	13.7	13.7		
	5% WB	MCDBR	17.5	25.4	29.1	34.6	38.4	33.1	28.2	30.4	34.0	28.7	16.1	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.311	0.361	0.405	0.413	0.407	0.404	0.388	0.352	0.339	0.311	0.281		
	taud		2.068	2.021	1.957	2.092	2.169	2.302	2.348	2.362	2.396	2.284	2.121	2.026		
	Ebn at Noon		219	246	255	261	266	269	267	266	263	237	202	189		
	Edh at Noon		24	35	46	45	44	39	37	35	30	27	23	21		
(44) All-Sky Solar Radiation	RadAvg		392	737	1217	1536	1740	1912	1736	1496	1120	672	361	266		
	RadStd		24	39	54	73	122	159	112	113	110	59	37	21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page